Experiment No. 6: Implementation of Multi-Variate Linear Regression using Python.

Aim:

To implement Multi-Variate Linear Regression using Python and analyze the relationship between one dependent variable and multiple independent variables.

Theory:

Multi-Variate Linear Regression is a supervised machine learning technique used to predict a dependent variable based on two or more independent variables. It extends simple linear regression by considering multiple features that influence the outcome. The relationship between variables is expressed using the equation:

$$Y = b_0 + b_1X_1 + b_2X_2 + ... + b_nX_n$$

Where Y is the dependent variable, x1,x2,xn… are independent variables, b0 is the intercept, and b1,b2,bn... are coefficients. In Python, libraries such as **NumPy, Pandas, Matplotlib, and Scikit-learn** are commonly used to perform regression analysis and visualize results.

Procedure:

1. Import required libraries such as Pandas, NumPy, and Scikit-learn.
2. Load the dataset containing multiple independent variables and one dependent variable.
3. Separate the dataset into independent variables (X) and dependent variable (Y).

4. Split the dataset into training and testing sets.
5. Create a linear regression model using the Scikit-learn library.
6. Train the model using the training dataset.
7. Predict the results using the testing dataset.
8. Evaluate the performance of the model using metrics such as Mean Squared Error or R^2 score.

Conclusion:

Thus, Multi-Variate Linear Regression was successfully implemented using Python. The model helps in predicting a dependent variable based on multiple independent variables and provides insights into the relationship between different factors affecting the outcome.